

Seoyong Nam

Ph.D. Student · Haptics & Multimodal AI

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Education

Hanyang University — Ph.D. Student, Applied Artificial Intelligence <i>Advised by Prof. Yongjae Yoo (MILab)</i> GPA 4.42 / 4.5	2025 – Present Seoul, Korea
Hanyang University — B.S. in Computer Science <i>Advised by Prof. Yongjae Yoo</i> GPA 3.72 / 4.5	2021 – 2025 Ansan, Korea

Research Interests

Haptics and multimodal interaction; automatic 4D / multisensory effect generation and rendering; multimedia AI and computer vision; accessibility; robot AI (vision–language–action and video world models).

Research Experience

Ph.D. Research Student — Multimodal Intelligence and Interaction Lab (MILab) Present <i>Hanyang University</i>	2025 – Present Ansan, Korea
- Automatic multimodal 4D effect generation and haptic rendering for immersive media. - Direct haptic rendering of Gaussian Splatting scenes; VLA and video world models for robot manipulation.	
Undergraduate Research Intern — MILab <i>Hanyang University</i>	2023 – 2025 Ansan, Korea
Audiovisual-to-tactile translation and automatic 4D effect authoring.	

Teaching Experience

Teaching Assistant, Hanyang University
2025 Spring — Linear Algebra (MAT2003); Speech Recognition (DAI3005) 2025 Fall — Numerical Analysis (MAT3008); Probability & Stochastic Processes (ELE3037) 2026 Spring — Tensor Programming (DAI2001); Computer Architecture (ENE1004); Speech Recognition (DAI3005)

Publications

Underlined name indicates first authorship.

International

- Nam, S.**, Jung, H., Yoo, Y. “HAPhy: AI-Assisted Content Creator Supporting Tools for Multisensory Haptic Experiences in Immersive Media.” *ACM Multimedia Systems Conference (MMSys)*, 2026.
- Chung, M., **Nam, S.**, Park, J., Yoo, Y. “TaME4D: Tailoring Multisensory Effects for Improving 4D Movie Watching Experiences.” *ACM International Conference on Multimedia (ACM MM)*, 2026.
- Nam, S.**, Yoo, Y. “Evaluating Perceptual Characteristics of Impact Haptic Stimuli.” *Extended Abstracts, IEEE World Haptics Conference (WHC)*, 2025.

4. Jung, H., **Nam, S.**, Yoo, Y. “Generative AI-Based Multimodal Authoring Tool for Virtual Objects.” *Demonstration, IEEE World Haptics Conference (WHC)*, 2025.
5. **Nam, S.**, Chung, M., Kim, H., Kim, E., Kim, T., Yoo, Y. “Automatic Generation of Multimodal 4D Effects for Immersive Video Watching Experiences.” *SIGGRAPH Asia Technical Communications*, 2024. (Oral)

Domestic (Korea)

1. **Nam, S.**, Yoo, Y. “A Preliminary Study Toward Direct Haptic Rendering of Gaussian Splatting Scenes: Deformable Object Rendering and Passivity Instrumentation on a Rendered Point Cloud.” *Korea Haptics Conference*, 2026. (Poster)
2. **Nam, S.**, Yoo, Y. “Direct Haptic Rendering of Gaussian Splatting Scenes: A Demonstration of Deformable Object Rendering on a Rendered Point Cloud.” *Korea Haptics Conference*, 2026. (Demo)
3. Kim, E., Park, J., Kim, S., **Nam, S.**, Yoo, Y. “Real-time AI-based Audio-Tactile Chart Accessibility System for the Visually Impaired.” *HCI Korea*, 2026.
4. **Nam, S.**, Park, J., Chung, M., Yoo, Y. “Audiovisual Feature Extraction for Multimodal Haptic Rendering in 4D Films Based on User Preferences.” *Korea Computer Congress (KCC)*, 2025.
5. Chung, M., **Nam, S.**, Yoo, Y. “Design of a Four-Directional Wind Feedback System Using Video-based Object Tracking.” *Korea Computer Congress (KCC)*, 2025.
6. Chung, M., **Nam, S.**, Yoo, Y. “Automatic Authoring of Temperature Feedback in 4D Films Using Retrieval-Augmented Generation.” *HCI Korea*, 2025.
7. **Nam, S.**, Kim, T., Chung, M., Yoo, Y. “Adaptive Threshold-Based Event Detection and Visual Effect Extraction for Automatic 4D Effect Authoring.” *Korea Computer Congress (KCC)*, 2024.
8. Kim, T., Jung, J., **Nam, S.**, Yoo, Y. “Speech-Gesture Recognition Model for Intuitive Control of Rail Robots.” *Korea Robotics Society Annual Conference (KRoC)*, 2024. (Poster)
9. **Nam, S.**, Chung, M., Kim, T., Yoo, Y. “Automated Generation of Multimodal 4D Effects in Home Theater Environments.” *Korea Haptics Conference*, 2023. (Demo)

Honors & Awards

- **2025** Student Innovation Challenge (Mixed Reality), Finalist — IEEE World Haptics 2025
- **2025** Best Work-in-Progress Paper Award, Finalist — World Haptics 2025
- **2025** Kakao × KIISE AI Agent Competition
- **2024** Grand Prize, SW/ICT/AI Conference — Hanyang University ERICA
- **2024** Student Idea Competition (Future AI Mobility), Finalist — Korean Haptics Society
- **2023** Grand Prize, Future Mobility Haptic Technology Idea Competition — Hyundai Motor & Korean Haptics Society
- **2023–2024** College of Computing Leadership Scholarship — Hanyang University ERICA

Skills

Programming: Python, PyTorch

Areas: Haptics, Multimodal Interaction, Computer Vision, Multimedia AI, Robot AI

Languages: Korean (native), English (professional working proficiency)